Exercise 7 – Bits & Bytes

1 bit	= 0 or 1
1 nibble	= 4 bits
1 byte	= 2 nibbles = 8 bits
1 word	= 2 bytes = 16 bits

2^{10} bytes	= 1024 bytes = 1 KB (kilo bytes)
2^{20} bytes	= 1024 KB = 1 MB (mega bytes)
2^{30} bytes	= 1024 MB = 1 GB (giga bytes)
2^{40} bytes	= 1024 GB = 1 TB (tera bytes)
2^{50} bytes	= 1024 TB = 1 PB (peta bytes)
2^{60} bytes	= 1024 PB = 1 EB (exa bytes)
2^{70} bytes	= 1024 EB = 1 ZB (zeta bytes)

Value range of an **n-bit** binary number = 2^n (0 to 2^n-1)

1-bit range = 2^1	(0 to 2^1-1)	=> values => 2
2-bit range = 2^2	(0 to 2^2-1)	=> values => 4
4-bit range = 2^4	(0 to 2^4-1)	=> values => 16
7-bit range = 2^7	(0 to 2^7-1)	=> values => 128
8-bit range = 2^8	(0 to 2^8-1)	=> values => 256
10-bit range = 2^{10}	(0 to 2^{10}-1)	=> values => 1024 = **1 KB**
16-bit range = 2^{16}	(0 to 2^{16}-1)	=> values => 65536 = **64 KB**
32-bit range = 2^{32}	(0 to 2^{32}-1)	=> values => 4,29,49,67,296 = **4 GB**
48-bit range = 2^{48}	(0 to 2^{48}-1)	=> values => 28,14,74,97,67,10,656 = **256 PB**
64-bit range = 2^{64}	(0 to 2^{64}-1)	=> values => 1,84,46,74,40,73,70,95,51,616 = **16 EB**

Question:

2,05,824 bytes	= 201 x 1024 bytes = 201 KB
3,98,45,888 bytes	= (?) MB
56 PB	= (?) GB
10,65,15,18,89,408 bytes	= (?) GB

Question:

8 bytes	= (?) bits
2 bytes	= (?) nibbles
4 words	= (?) bytes

Question:

2^{12} bytes	= $2^{(2+10)} = 2^2 \times 2^{10} = 2^2 \times$ 1KB = 4 KB
2^{18} bytes	= (?) KB
2^{23} bytes	= (?) MB
2^{35} bytes	= (?) GB

Question:

How many values can be represented using **5-bits**?
What is the most significant bit of $(1100)_2$?
What is the least significant byte of $(0110110101011101)_2$?
What is the most significant nibble of $(00011110)_2$?
What is the decimal value of $(10000000000)_2$?